Norfolk Southern (NSC)

The Merger Arbitrage Mirage Obscures Severe Cost Inflation and Cash Depletion

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> Norfolk Southern trades at an extraordinary disconnect from underlying operational cash flow, buoyed by the pending Union Pacific transaction while core rail operations battle aggressive cost headwinds. While top-line intermodal volumes showed recovery in recent quarters, operating ratio deterioration, surging fuel expense, and significant working capital outflows have severely compressed free cash conversion. The equity market prices in an idealized transaction execution that ignores both the regulatory absorption timeline and the fundamental erosion of cash generation. We establish a probability-weighted blended target of $184.66, representing -43.4% upside versus the reference price of $326.40.

HOLD	$184.66	-43.4%	$326.40
Hold — Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

The market currently prices Norfolk Southern as a clean regulatory arbitrage conduit, looking past sequential operating cash flow contraction and persistent non-operating transaction drag. Our analysis indicates that mounting fuel inflation, purchased services growth, and elevated capital expenditures leave core rail operations fundamentally overvalued relative to realized cash conversion.

EXECUTIVE SUMMARY

Norfolk Southern operates an essential freight rail network traversing the Eastern United States, moving intermodal containers, automotive shipments, chemicals, and coal across the industrial Midwest and Southeast. The franchise occupies an established duopoly across eastern corridors, acting as a critical logistical backbone for domestic supply chains and manufacturing centers. In July 2025, the company agreed to be acquired by Union Pacific, initiating an extensive regulatory review process. Throughout recent quarters, transaction-related expenditures and operational adjustments linked to the pending merger have consistently burdened reported earnings, creating a wedge between statutory GAAP performance and underlying freight operations.

Our review of the evidentiary record reveals a company navigating acute operational crosscurrents. The second quarter of 2026 delivered an all-time record revenue of $3.5B, rising 11% year over year on a 4% volume expansion and a 22% surge in intermodal activity. However, profitability failed to follow top-line growth. Net income contracted 4% to $734M as operating expenses surged 21% to $2.34B, heavily penalized by an 85% spike in fuel costs to

$405M. The quarterly operating ratio deteriorated to 67.6%, with the adjusted operating ratio climbing 210 basis points to 65.5%. More concerning for equity holders, first-half operating cash flow collapsed 31% to $1.39B under severe working capital strain, leaving free cash flow at approximately $577M, an amount barely sufficient to support dividend distributions.

This follows an equally challenging start to the year. In the first quarter of 2026, revenue stood flat at $3.0B amid a 1% volume contraction, while GAAP net income fell 27% to $547M due to merger friction, fuel costs, and harsh weather. Adjusted operating ratio expanded 80 basis points to 68.7%, and operating cash flow declined to $344M compared to $950M in the prior-year period. Management curtailed capital expenditures by approximately $300M, or 14%, to defend free cash flow while upholding a $150M cost-savings commitment. Full-year FY2025 demonstrated operational resilience, delivering $12.2B in revenue, $4.57B in EBIT, and diluted EPS of $12.75, assisted by more than $215M in productivity savings. Nevertheless, recent quarters confirm that expense growth is outstripping top-line gains.

Capital structure discipline remains compromised by debt burdens and reduced balance sheet liquidity. Norfolk Southern closed FY2025 with $17.3B in total debt and $1.53B in cash reserves, yielding net debt of $15.8B. Enterprise value stands at $89.1B against a market capitalization of $73.3B at the reference price of $326.40. Valuation multiples reflect substantial optimism, with trading levels establishing a P/E of 25.6x, an EV to EBITDA multiple of 15.2x, and an EV to revenue multiple of 7.3x. Given our live discount rate of 10.03% and three rigorously modeled cash flow paths, the current equity pricing lacks structural support, reinforcing our Hold rating based on clear valuation concerns.

OPERATIONAL SNAPSHOT

$326.40	$12.2B	$2.16B	17.7%	1.29	$326.40
REF. PRICE	FY2025 REVENUE	FY2025 FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-20.

THE MOAT IS STRUCTURAL AND ENDURING

The economic moat protecting Norfolk Southern stems from irreplaceable linear infrastructure across the Eastern United States. Replicating thousands of route miles connecting ports, industrial hubs, chemical refineries, and population centers across the Midwest and Southeast is economically impossible due to prohibitive land assembly expenses and environmental permitting hurdles. Shippers operating heavy manufacturing, chemical production, and automotive assembly facilities design their logistical footprints around rail sidings, locking in long-term operational reliance. This physical integration creates elevated switching costs, as diverting freight from rail to highway trucking involves substantial cost premiums, driver availability constraints, and significant logistical inefficiency.

Pricing power across core merchandise categories remains durable despite cyclical volume softness. In the second quarter of 2026, merchandise revenue expanded 8.1% to $2.13B, supported by core pricing gains even amidst

broader freight volatility. Similarly, during the first quarter of 2026, merchandise served as an anchor, expanding 1.2% to $1.88B despite negative overall volume growth. Customers transporting bulky, hazardous, or high-volume industrial goods have limited transportation alternatives, enabling Norfolk Southern to pass through baseline contract increases over multi-year cycles. This foundational pricing leverage forms the core defense against broader inflation.

However, the franchise remains exposed to structural shifts in intermodal and bulk handling. While intermodal revenue surged 22% to $908M in the second quarter of 2026, driven by highway-to-rail conversions, the franchise experienced competitive market-share loss in prior periods. Specifically, in the third quarter of 2025, intermodal volumes slipped 2% as the collaborative alliance between BNSF and CSX contested freight corridors in the Southeast. Furthermore, seaborne coal pricing weakness contributed to volume drops of 5% in that period and 11% in the fourth quarter of 2025. The moat protects baseline network utilization, but maintaining volume share across competitive corridors requires continuous operational efficiency and consistent service performance.

Regulatory scrutiny surrounding the pending acquisition by Union Pacific introduces ongoing uncertainty regarding network commitments and commercial autonomy. The merger announcement in July 2025 initiated intense regulatory review that continues to create frictional costs across corporate and network operations. Until regulatory authorities reach a definitive determination, Norfolk Southern must operate within an operational holding pattern, balancing productivity goals against transaction integration requirements. This regulatory oversight constrains aggressive network restructuring while keeping operating expenses elevated.

VALUATION

Three scenarios make the valuation range explicit

Our valuation framework evaluates Norfolk Southern through an explicit multi-scenario discounted cash flow model covering FY2026 through FY2030, applying a live WACC of 10.03% and an equity beta of 1.29 across 224.6 million diluted shares. The prevailing market price of $326.40, representing an aggregate market capitalization of $73.3B, embeds operational expectations that significantly exceed historical cash flow conversion and achievable revenue expansion. Market participants appear to be treating the Union Pacific acquisition as fully secured without discounting potential regulatory delays, integration complications, or intermediate cash flow degradation.

Across our forecast window, revenue growth and normalized free cash flow margins anchor the three fundamental scenarios. Terminal value calculations employ Gordon growth mechanics driven by our 10.03% discount rate. In our baseline projection, top-line compounding stabilizes at a 3.8% CAGR while target free cash flow margin reaches 21.2% by FY2030. Under more adverse operating conditions, revenue compounds at 2.7% with free cash flow margin constrained to 18.5%. Conversely, our constructive scenario models revenue compounding at 4.8% with free cash flow margins expanding to 23.0% as productivity initiatives take hold.

Probability weighting these explicit scenarios at 25% bear, 50% base, and 25% bull produces a blended equity value of $184.66 per share. This outcome demonstrates that current equity prices are detached from fundamental cash flow generation across every evaluated scenario. Even under our most optimistic bull case, which yields a fair value of $228.57, the shares face 30.0% downside. The base case target of $193.26 indicates a 40.8% deficit, while the bear case indicates potential erosion of 62.1% toward $123.57.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	2.7%	18.5%	$123.57	-62.1%
Base	50%	3.8%	21.2%	$193.26	-40.8%
Bull	25%	4.8%	23.0%	$228.57	-30.0%

Bear $123.57

Severe cost inflation persists as fuel expenses and purchased services outstrip freight pricing. Regulatory challenges or restructuring costs around the Union Pacific transaction permanently compress free cash flow margins to 18.5%, while revenue growth languishes at a 2.7% CAGR. Operating cash flow fails to rebound from working capital drains, leaving the valuation impaired at $123.57 per share.

Base $193.26

Core merchandise pricing power stabilizes top-line expansion at a 3.8% CAGR while intermodal volumes normalize after initial conversion spikes. Operating ratios recover modestly toward historical ranges, enabling target free cash flow margins to settle at 21.2%. Balanced cash conversion against ongoing network capital expenditures establishes an intrinsic value of $193.26 per share.

Bull $228.57

Highway-to-rail intermodal conversion accelerates across the Southeast corridor, supporting a 4.8% revenue CAGR. Productivity targets exceed projections, fully absorbing fuel and labor inflation to drive target free cash flow margins to 23.0%. Disciplined capital expenditures and working capital stabilization generate our bull target of $228.57 per share.

VALUATION FRAMEWORK

A live discount rate, not a static hurdle

Our quantitative framework deploys an explicit weighted average cost of capital of 10.03%, reflecting Norfolk Southern's balance sheet structure and operating exposure. Incorporating an equity beta of 1.29 captures the cyclical volatility inherent in freight rail volumes, seaborne coal fluctuations, and macro-industrial production across the Midwest and Southeast. The capital structure carries $17.3B in total debt and $15.8B in net debt against total equity market capitalization of $73.3B, requiring a rigorous cost of capital that accounts for financial leverage and corporate transaction risk.

Market pricing at $326.40 implies an unrealistically depressed cost of capital or an assumption of immediate cash flow parity with acquisition terms. By applying our 10.03% hurdle rate, we ensure that cash flows generated across the FY2026 to FY2030 forecast window reflect genuine economic returns. Norfolk Southern operates in an asset-intensive sector subject to periodic operational bottlenecks,

KEY VALUATION PARAMETERS

Forecast window	**FY2026 to FY2030**
Live WACC	**10.03% (2026-10-20)**
Equity beta	**1.29**
Diluted shares	**224.6 million**

HOLD

over $184.66 blended target

safety outlays, and fuel price shocks. Our discount rate appropriately penalizes these operational disruptions rather than presuming frictionless multi-year compounding.

FCF margin normalization is the engine

Cash conversion dynamics serve as the central driver of our valuation differential. In FY2025, Norfolk Southern converted $12.2B in revenue into $2.16B in free cash flow, representing an FCF margin of 17.7%. Recent quarters, however, illustrate substantial volatility in this metric. During the first half of 2026, free cash flow approximately halved to roughly $577M as operating cash flow contracted 31% to $1.39B, damaged by working capital outflows of $388M in the first quarter alone. Capital expenditure curtailments of roughly $300M, or 14%, were deployed in early 2026 specifically to shield cash reserves.

Our forecast scenarios model an eventual structural recovery in cash conversion above FY2025 levels, reaching terminal margins of 18.5% in the bear case, 21.2% in the base case, and 23.0% in the bull case. Achieving the base target of 21.2% requires management to deliver on its $150M cost-savings commitment for 2026, reverse the working capital leakages that plagued early 2026, and stabilize the operating ratio below recent levels of 67.6%. Because current spot pricing implies cash flow conversion well beyond our 23.0% bull scenario, valuation risk remains skewed to the downside.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

Methodology. Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

BUSINESS & SEGMENT DRIVERS

Revenue prospects anchor on Intermodal Conversions and Merchandise Pricing Power

HOLD

Norfolk Southern organizes its freight franchise across intermodal, automotive, chemicals, and coal segments, moving industrial inputs and finished goods across the Eastern rail network. Revenue trajectory over the medium term depends primarily on sustaining highway-to-rail conversions within intermodal corridors while exercising core

pricing power in the merchandise franchise. In the second quarter of 2026, intermodal revenue demonstrated dramatic momentum, surging 22% year over year to $908M, supported by broad 4% volume expansion across the enterprise. This performance marks a sharp reversal from late 2025, where intermodal volumes contracted 7% in the fourth quarter and 2% in the third quarter under aggressive competitive pressure from the BNSF-CSX alliance in the Southeast.

The merchandise franchise, encompassing chemicals, automotive, and industrial products, remains the foundation of rail profitability. In the second quarter of 2026, merchandise revenue expanded 8.1% to $2.13B, reflecting strong commercial pricing execution that offset inflationary pressures elsewhere in the network. During the first quarter of 2026, merchandise expanded 1.2% to $1.88B despite negative overall volume growth, cushioning operating earnings. Management relies on merchandise pricing strength to support network operating income while productivity initiatives work through railway operations.

Productivity execution has delivered tangible operational milestones, yet expense growth continues to challenge overall efficiency. Full-year FY2025 demonstrated the potential of these programs, capturing more than $215M in productivity savings and driving the operating ratio down 220 basis points to 64.2%. However, these efficiencies encountered substantial headwinds in 2026. In the second quarter of 2026, operating expenses jumped 21% to $2.34B against 11% top-line growth, driven by an 85% explosion in fuel costs to $405M. Consequently, the operating ratio expanded to 67.6%, with the adjusted metric worsening 210 basis points to 65.5%, demonstrating that fuel volatility and service expenses can rapidly overwhelm underlying operational gains.

Coal transportation remains an ongoing operational drag, constrained by global commodity trends and domestic utility consumption. In the third quarter of 2025, coal volumes dropped 5% on weak seaborne pricing, followed by an 11% volume decline in the fourth quarter of 2025. While merchandise and intermodal corridors have shown volume recovery, coal loadings continue to face cyclical and secular headwinds, limiting its contribution to overall freight density across eastern routes.

What to monitor

- **Fuel expense stabilization** represents the primary catalyst for operating ratio recovery following the 85% cost surge to $405M observed in the second quarter of 2026.
- **Intermodal corridor retention** against competing eastern rail alliances will determine whether second-quarter volume momentum of 22% top-line growth can be sustained long-term.
- **Working capital normalization** must be tracked closely after first-quarter cash outflows of $388M contributed to a 31% collapse in first-half operating cash flow.
- **Merger transaction expenses** linked to the Union Pacific acquisition must be monitored to evaluate continued drag on GAAP earnings and corporate resources.

+0.5%	36.3%	37.5%	17.7%	$15.8B	18.5%
FY2025 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET DEBT	RETURN ON EQUITY

METRIC	FY2025 (AS REPORTED)

Revenue	$12.2B
Gross Profit	$4.43B
EBIT	$4.57B
Net Income	$2.87B
EBITDA	$5.85B
Free Cash Flow	$2.16B
Diluted EPS	$12.75
ROE	18.5%

RISK & MOAT MATRIX

The bear case

SHARE EROSION

Competitive freight alliances between western and eastern operators, such as the BNSF-CSX partnership, continue to contest key Southeast intermodal volumes. Loss of high-density container traffic threatens to reverse recent highway

M&A DIGESTION

The pending acquisition by Union Pacific announced in July 2025 continues to generate substantial legal, regulatory, and restructuring charges that depress reported earnings. Extended regulatory review creates organizational distraction and delays critical capital investment

REGULATORY AND PRICING SCRUTINY

Federal oversight over rail service performance, safety compliance, and commercial demurrage charges limits unconstrained pricing flexibility across captive shipper segments. Elevated regulatory intervention risks capping core merchandise rate increases below network cost inflation.

TECHNOLOGY DISRUPTION

Delays in deploying advanced rail automation, digital dispatching, and fuel efficiency systems threaten network fluidness relative to regional competitors. Inability to meet productivity targets compromises long-term cost containment objectives across heavy locomotive maintenance.

REINVESTMENT TREADMILL

Maintaining extensive trackage, bridges, locomotives, and terminal facilities requires relentless capital deployment regardless of underlying traffic volume. Curtailing capital expenditures, as demonstrated by the $300M reduction in early 2026, cannot continue indefinitely without impairing network velocity.

WACC AND DURATION SENSITIVITY

Carrying $17.3B in total debt against volatile operational cash flows leaves equity valuation highly sensitive to capital market conditions under our 10.03% discount rate. Cash flow contractions directly heighten leverage risk, limiting capacity for shareholder distributions.

conversion gains and depress corridor density.

decisions across core rail operations.

The strengths

OLIGOPOLY POSITION

The permanent physical infrastructure network throughout the Eastern United States creates insurmountable barriers to entry against prospective rail entrants. Captive industrial manufacturing sidings guarantee essential baseline volume across long-term industrial cycles.

FCF COMPOUNDING

Historically demonstrated cash generation, highlighted by $2.16B in FY2025 free cash flow, confirms the underlying capacity of the asset base when operational disruptions normalize. Structured cost-reduction programs, including the $150M savings target for 2026, provide operating leverage.

GEOGRAPHIC DURABILITY

Direct connectivity across high-density manufacturing centers in the industrial Midwest and population centers in the Southeast supports broad industrial freight diversity. Critical corridor access insulates core merchandise segments from localized economic shocks.

PIPELINE OPTIONALITY

Highway-to-rail conversion trends offer multi-year volume expansion opportunities as motor carrier freight transitions onto fuel-efficient rail corridors. Continued corporate demand for low-carbon transportation supports long-term intermodal volume density.

The bottom line on risk and reward

Norfolk Southern presents a severe mismatch between market valuation and fundamental cash flow generation. At the prevailing spot price of $326.40, the equity trades at 25.6x FY2025 diluted earnings of $12.75, 6.0x revenue, and 15.2x EBITDA. These valuation multiples reflect an extraordinary premium for an enterprise operating in an asset-intensive, cyclically exposed industry currently burdened by elevated operating expenses and persistent transaction drag. While top-line records demonstrate that volume demand is improving, equity valuation must ultimately be governed by free cash flow conversion rather than top-line metrics alone.

Recent operational performance confirms that inflationary headwinds and working capital requirements are severely restricting cash availability. The first half of 2026 saw operating cash flow decline 31% to $1.39B, with free cash flow of approximately $577M barely covering dividend obligations. Although productivity programs delivered more than $215M in savings in FY2025 and targeted an additional $150M in 2026, recent operating expense inflation of 21% outpaced top-line growth of 11%, driving the operating ratio up to 67.6%. These metrics contradict the thesis that cash flow margins will expand rapidly in the near term.

Our probability-weighted valuation framework, incorporating a live 10.03% WACC across explicit DCF scenarios, yields a blended fair value of $184.66 per share, representing -43.4% upside versus the reference price. With the bear target at $123.57, base target at $193.26, and bull target at $228.57, current spot levels reside substantially above even our most constructive operating path. We advise institutional investors to maintain a disciplined stance, reiterating our Hold rating based on pronounced valuation concerns.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

FRANCHISE DEEP DIVE

Intermodal and Merchandise Freight: Volume Rebound Meets Cost Realities

Norfolk Southern's freight delivery engine centers on its intermodal and merchandise operations across the Eastern United States, moving consumer goods, automotive parts, chemicals, and industrial commodities. Intermodal operations represent the primary vector for volume growth, capturing freight from highway trucking through dedicated container terminals and collaborative logistics corridors. In the second quarter of 2026, this dynamic was vividly illustrated as intermodal revenue surged 22% year over year to $908M, propelling total corporate revenue to an all-time record of $3.5B. This turnaround confirmed that freight demand was rebounding following stagnation across FY2025.

However, running high-velocity intermodal networks introduces significant operating leverage that cuts both ways when input costs escalate. The second quarter of 2026 demonstrated this vulnerability, as railway operating expenses jumped 21% to $2.34B. The primary catalyst was an 85% surge in fuel expense, which reached $405M. Despite record volume and revenue, net income slipped 4% to $734M, and the operating ratio deteriorated to 67.6%, or 65.5% on an adjusted basis. The operational challenge lies in sustaining network velocity and equipment availability without permitting cost inflation to erode freight profitability.

The merchandise segment provides the structural margins required to absorb network volatility. Moving specialized chemical tank cars, raw industrial materials, and finished automobiles across the industrial Midwest and Southeast allows Norfolk Southern to command firm contract pricing. Merchandise revenue advanced 8.1% to $2.13B in the second quarter of 2026 and grew 1.2% to $1.88B during the adverse conditions of the first quarter of 2026, demonstrating commercial resilience. Nonetheless, network performance must balance merchandise margin generation against the transaction-related disruptions associated with the Union Pacific merger announced in July 2025.

What the recent record shows

- Intermodal revenue expanded 22% year over year to $908M in the second quarter of 2026, driving all-time record quarterly revenue of $3.5B on 4% total volume expansion.
- Operating expenses escalated 21% to $2.34B in the second quarter of 2026, driven by an 85% increase in fuel expense to $405M and worsening the operating ratio to 67.6%.
- First-half 2026 operating cash flow contracted 31% to $1.39B, resulting in free cash flow of approximately $577M that barely covered dividend commitments.
- First-quarter 2026 operating cash flow dropped to $344M from $950M in the prior year, impaired by a $388M working capital drain that prompted a $300M capital expenditure curtailment.

Our read: Norfolk Southern's freight network possesses undeniable physical utility and demonstrated pricing power across merchandise categories, but recent performance proves that volume gains are currently

overwhelmed by fuel expense and working capital drain. At $326.40 per share, the market has detached from underlying cash generation, fully justifying our Hold stance and $184.66 blended valuation.